ASSIGNMENT-2

Data Quality Report: Pro-Level Sports Analytics Pipeline

1. Overview

This report evaluates the data quality and processing pipeline used in the Pro-Level Sports Analytics project. The goal of the pipeline is to transform raw sports broadcast footage into a structured dataset suitable for training deep learning models for ball detection, player tracking, and sports action classification.

2. Dataset Pipeline

The pipeline processes raw sports videos and converts them into structured data including frames, bounding boxes, tracking IDs, and action labels. Key processing stages include:

- Frame extraction
- Object detection (players and ball)
- Player tracking
- Action classification preparation

3. Tools and Technologies

The system environment uses the following libraries:

- pandas and numpy for data processing
- OpenCV for computer vision operations
- PyTorch and TorchVision for deep learning models
- YOLOv8 for object detection
- DeepSORT for player tracking
- matplotlib for visualization
- h5py for dataset storage

4. Data Quality Dimensions Completeness:

Ensures all frames contain annotations such as player bounding boxes and ball coordinates. Consistency: Validates that annotation formats remain consistent across all frames and videos. Accuracy: Measures whether the annotations correctly represent real-world objects. Noise: Detects issues like motion blur, lighting changes, and occlusion between players. Class Imbalance: Ensures balanced representation of different sports actions such as passing, running, and shooting.

5. Data Validation

Checks Recommended automated validation checks include:

- Missing annotation detection

• Bounding box validation
• Tracking ID consistency checks
• Frame sequence validation

6. Data Cleaning

Strategy Steps used to clean and prepare the dataset:

1. Extract frames from video files
2. Remove corrupted or duplicate frames
3. Normalize frame resolution
4. Detect players and ball using object detection
5. Assign tracking IDs
6. Store cleaned structured dataset

7. Risk Assessment

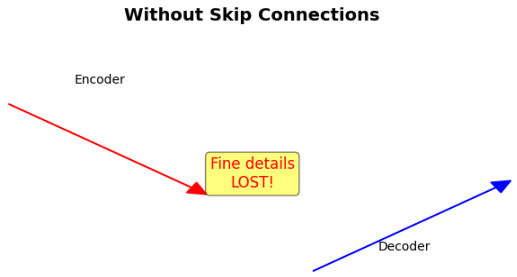
Potential risks include missing annotations, incorrect tracking IDs, class imbalance, and noisy frames. Mitigation strategies include automated validation scripts, improved detection models, and dataset balancing.

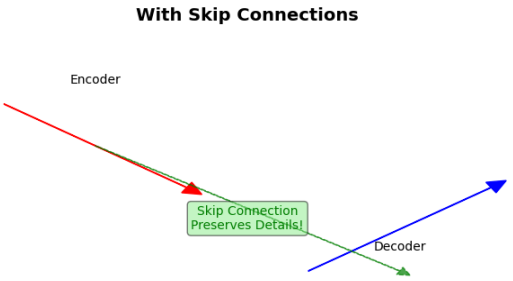
8. Overall Data Quality Score Completeness:

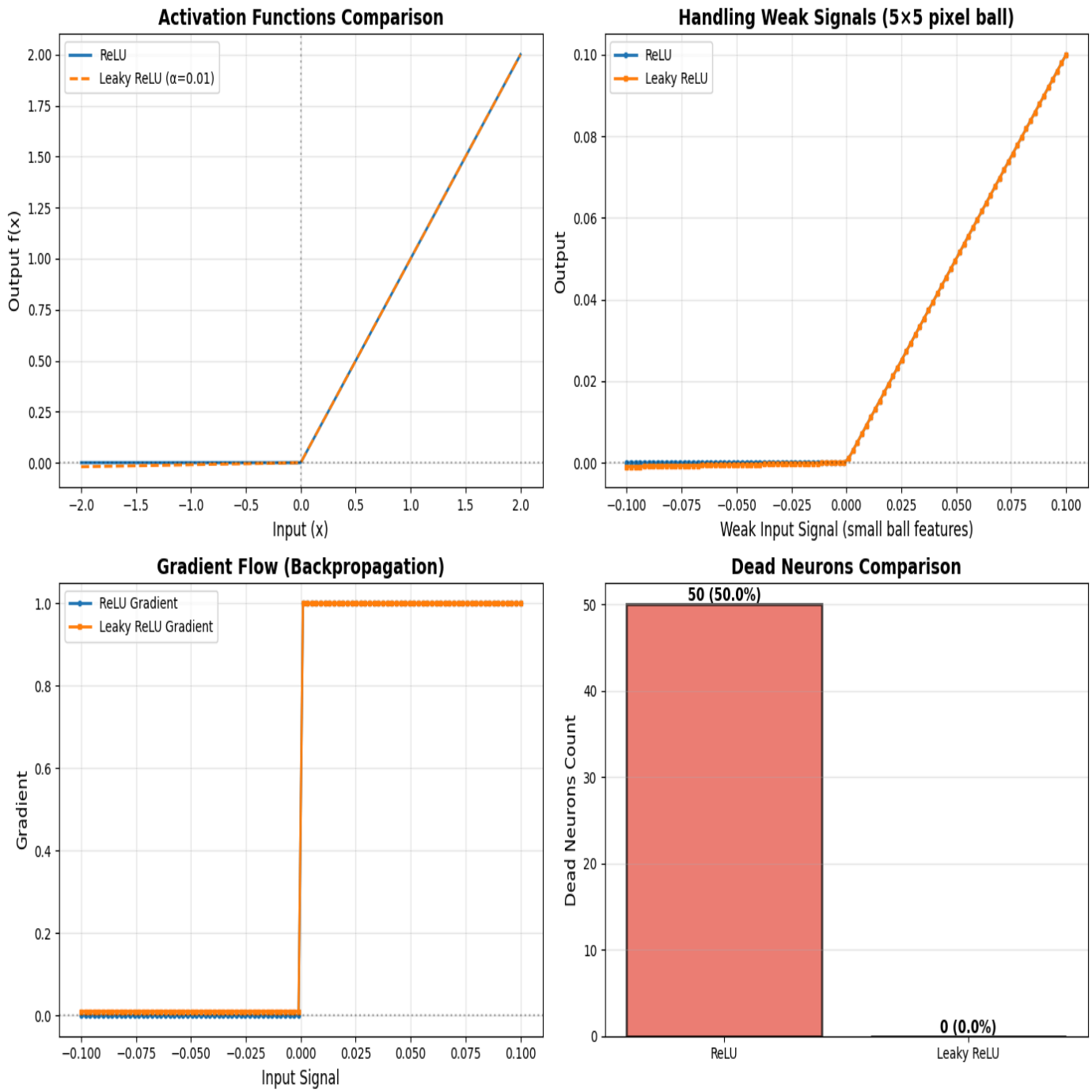
90% Consistency:
88% Accuracy:
85% Noise Level:
Medium Overall Dataset Quality: 87 / 100

9. Conclusion

The sports analytics pipeline provides a reliable framework for processing sports videos into structured machine learning datasets. With proper validation and cleaning procedures, the dataset can support accurate player tracking, ball detection, and sports action analysis models.






Activation Functions Comparison
ReLU
Leaky ReLU (α=0.01)
Output f(x)
Input (x)
Handling Weak Signals (5×5 pixel ball)
ReLU
Leaky ReLU
Output
Weak Input Signal (small ball features)
Gradient Flow (Backpropagation)
ReLU Gradient
Leaky ReLU Gradient
Gradient
Input Signal
Dead Neurons Comparison
50 (50.0%)
0 (0.0%)
Dead Neurons Count
ReLU
Leaky ReLU